

END-SEMESTER MOCK TEST

Full Practice Paper — All Topics

Total Questions: 60	Duration: 2 Hours	Marks: 60 (1 per Q)	No Negative Marking
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Section	Topics Covered	Questions
Section 1: Quant & DI	Ratio & Partnership Speed, Time & Trains Algebra Mensuration	Q1 - Q20
Section 2: Reasoning	Blood Relations Direction Sense Syllogism Seating Arrangement Coding-Decoding	Q21 - Q40
Section 3: Verbal	Reading Comprehension Error Detection Vocabulary Fill in the Blanks	Q41 - Q60

Instructions: Each question carries 1 mark. Attempt all questions. Calculators not allowed. Circle the correct option on your answer sheet.

SECTION 1: QUANTITATIVE APTITUDE & DATA INTERPRETATION (Q1-Q20)

Topics: Ratio & Partnership | Speed, Time & Trains | Algebra Equations | Mensuration

RATIO & PARTNERSHIP (Q1-Q5)

Q1. A invests Rs. 5,000 for 12 months, B invests Rs. 8,000 for 6 months and C invests Rs. 3,000 for 9 months in a business. The total profit at the end of the year is Rs. 19,800. Find B's share of the profit.

- (A) Rs. 6,600
- (B) Rs. 7,040
- (C) Rs. 5,940
- (D) Rs. 8,250

Q2. A and B invest in a business in the ratio 4:5. A invests for 8 months and B invests for 6 months. If A's share of profit is Rs. 4,800, find B's share.

- (A) Rs. 4,500
- (B) Rs. 5,000
- (C) Rs. 4,800
- (D) Rs. 5,400

Q3. A mixture of 60 litres contains milk and water in ratio 2:1. How many litres of water must be added so that the ratio becomes 1:2?

- (A) 40 litres
- (B) 45 litres
- (C) 60 litres
- (D) 80 litres

Q4. A starts a business with Rs. 40,000. After 3 months, B joins with Rs. 60,000. After another 3 months, C joins with Rs. 80,000. At end of year, total profit is Rs. 1,68,000. Find C's share.

- (A) Rs. 48,000
- (B) Rs. 53,760
- (C) Rs. 56,000
- (D) Rs. 64,000

Q5. Two numbers are in ratio 3:5. If 9 is subtracted from each, the new ratio becomes 12:23. Find the larger number.

- (A) 45
- (B) 50
- (C) 51
- (D) 55

SPEED, TIME, DISTANCE & TRAINS (Q6-Q10)

Q6. A train 150m long passes a man running at 9 km/h in the same direction in 15 seconds. Find the speed of the train.

- (A) 36 km/h
- (B) 40 km/h
- (C) 45 km/h
- (D) 54 km/h

Q7. Two trains 120m and 180m long are running in opposite directions at 50 km/h and 40 km/h respectively. How long will they take to cross each other?

- (A) 10 seconds
- (B) 12 seconds
- (C) 15 seconds
- (D) 18 seconds

Q8. A man walks from A to B at 4 km/h and returns from B to A at 6 km/h. Find his average speed for the entire journey.

- (A) 4.8 km/h
- (B) 5 km/h
- (C) 5.2 km/h
- (D) 4.5 km/h

Q9. A car covers the first half of a journey at 40 km/h and the second half at 60 km/h. Find the average speed for the entire journey.

- (A) 50 km/h
- (B) 52 km/h
- (C) 48 km/h
- (D) 46 km/h

Q10. A thief steals a car at 2:00 PM and drives at 40 km/h. The police start chasing at 3:00 PM at 60 km/h. At what time will the police catch the thief, and how far from the starting point?

- (A) 4:00 PM, 80 km
- (B) 5:00 PM, 120 km
- (C) 5:30 PM, 140 km
- (D) 6:00 PM, 160 km

ALGEBRA EQUATIONS (Q11-Q15)

Q11. If $x + 1/x = 6$, find the value of $x^3 + 1/x^3$.

- (A) 180
- (B) 192
- (C) 198
- (D) 210

Q12. Find the sum and product of roots of the equation $3x^2 - 8x + 5 = 0$.

- (A) Sum = $8/3$, Product = $5/3$
- (B) Sum = $-8/3$, Product = $5/3$
- (C) Sum = $8/3$, Product = $-5/3$
- (D) Sum = $3/8$, Product = $3/5$

Q13. For what value of k does the equation $kx^2 - 6x + 1 = 0$ have equal roots?

- (A) 6
- (B) 7
- (C) 9
- (D) 12

Q14. A student copied the constant term incorrectly as 16 (instead of 6) and obtained roots 2 and 8. Another student copied the x -coefficient wrong as -5 (instead of -7) and obtained roots 1 and 6. Find the correct quadratic equation.

- (A) $x^2 - 10x + 6 = 0$
- (B) $x^2 - 10x + 16 = 0$
- (C) $x^2 - 6x + 6 = 0$
- (D) $x^2 - 11x + 6 = 0$

Q15. If α and β are roots of $x^2 - 5x + 6 = 0$, form the equation whose roots are $2\alpha + 1$ and $2\beta + 1$.

- (A) $x^2 - 12x + 35 = 0$
- (B) $x^2 - 11x + 35 = 0$
- (C) $x^2 - 12x + 28 = 0$
- (D) $x^2 - 10x + 35 = 0$

MENSURATION (Q16-Q20)

Q16. A rectangle is 30m long and 20m wide. A path of width 2m runs along the inside of all four sides. Find the area of the path.

- (A) 200 m²
- (B) 176 m²
- (C) 184 m²
- (D) 192 m²

Q17. A metallic sphere of radius 6 cm is melted and recast into small spheres each of radius 2 cm. How many small spheres are formed?

- (A) 18
- (B) 24
- (C) 27
- (D) 36

Q18. Find the area of a triangle with sides 13 cm, 14 cm and 15 cm using Heron's formula.

- (A) 72 sq cm

- (B) 84 sq cm
- (C) 90 sq cm
- (D) 96 sq cm

Q19. A cylinder and a cone have the same base radius and the same height. If the volume of the cylinder is 462 cm^3 , find the volume of the cone.

- (A) 154 cm^3
- (B) 231 cm^3
- (C) 308 cm^3
- (D) 77 cm^3

Q20. The length of a rectangle is increased by 20% and the breadth is decreased by 10%. What is the percentage change in the area?

- (A) 10% increase
- (B) 12% increase
- (C) 8% increase
- (D) 8% decrease

SECTION 2: REASONING (Q21-Q40)

Topics: Blood Relations | Direction Sense | Syllogism | Seating Arrangement | Coding-Decoding

BLOOD RELATIONS (Q21-Q24)

Q21. Pointing to a woman, Raj says, "She is the daughter of the only son of my grandfather."

How is the woman related to Raj?

- (A) Daughter
- (B) Sister
- (C) Niece
- (D) Cousin

Q22. A is B's brother. C is A's mother. D is C's father. E is D's wife. How is A related to E?

- (A) Son
- (B) Grandson
- (C) Nephew
- (D) Great-grandson

Q23. Introducing a man, a woman says: "His mother is the only daughter of my mother."

How is the woman related to the man?

- (A) Mother
- (B) Sister
- (C) Grandmother
- (D) Aunt

Q24. In a coded family language: $P + Q$ means P is the father of Q; $P - Q$ means P is the sister of Q; $P \times Q$ means P is the brother of Q; $P \div Q$ means P is the mother of Q. In the expression $A + B - C \times D$, how is A related to D?

- (A) Uncle
- (B) Grandfather
- (C) Father
- (D) Brother

DIRECTION SENSE (Q25-Q28)

Q25. A man starts from point P, walks 5 km North, then turns East and walks 3 km, then turns South and walks 5 km. How far is he from P and in which direction?

- (A) 5 km East
- (B) 3 km East
- (C) 3 km West
- (D) 8 km South

Q26. Ramesh starts from his home, walks 4 km East, then 3 km North, then 4 km West. How far and in what direction is he from home?

- (A) 3 km North
- (B) 4 km East
- (C) 3 km South
- (D) 7 km North

Q27. Priya starts from her office, walks 6 km South, then turns right and walks 4 km, then turns right and walks 6 km. How far and in which direction is she from her office?

- (A) 4 km East
- (B) 4 km West
- (C) 6 km North
- (D) 4 km North

Q28. A and B start from the same point. A walks 5 km North then 4 km East. B walks 4 km East then 3 km South. What is the shortest distance between A and B?

- (A) 5 km
- (B) 6 km
- (C) 8 km
- (D) 10 km

SYLLOGISM (Q29-Q33)

Q29. Statements: All birds are animals. No animal is a plant. Conclusions: I. No bird is a plant. II. Some birds are plants.

- (A) Only I follows
- (B) Only II follows
- (C) Both I and II follow
- (D) Neither I nor II follows

Q30. Statements: Some books are pens. All pens are pencils. Conclusions: I. Some books are pencils. II. All pencils are books.

- (A) Only I follows
- (B) Only II follows
- (C) Both I and II follow
- (D) Neither I nor II follows

Q31. Statements: All trees are flowers. Some flowers are fruits. Conclusions: I. Some trees are fruits. II. Some fruits are trees.

- (A) Only I follows
- (B) Only II follows
- (C) Both I and II follow
- (D) Neither I nor II follows

Q32. Statements: No cat is a dog. Some dogs are animals. Conclusions: I. No cat is an animal. II. Some animals are not cats.

- (A) Only I follows
- (B) Only II follows
- (C) Both I and II follow
- (D) Neither I nor II follows

Q33. Statements: All mobiles are phones. All phones are gadgets. Some gadgets are electronics. Conclusions: I. All mobiles are gadgets. II. Some electronics are gadgets. III. Some gadgets are mobiles.

- (A) Only I and II follow
- (B) Only II and III follow
- (C) Only I and III follow
- (D) All I, II and III follow

SEATING ARRANGEMENT (Q34-Q37)

Directions (Q34-Q37): Read the information carefully and answer the questions.

Six friends A, B, C, D, E and F are seated in a straight row facing North. The following conditions apply: (i) A is at the leftmost position. (ii) F sits immediately to the right of A. (iii) B is at the rightmost position. (iv) E sits immediately to the left of B. (v) D is 3rd from the right end. (vi) C and B are not adjacent.

Q34. Who is sitting immediately to the right of C?

- (A) E
- (B) F
- (C) D
- (D) A

Q35. Who is sitting 4th from the right end?

- (A) F
- (B) C
- (C) D
- (D) A

Q36. How many persons are sitting between A and E?

- (A) 2
- (B) 3
- (C) 4
- (D) 1

Q37. Which of the following pairs are sitting at the two ends of the row?

- (A) A and E
- (B) F and B
- (C) A and D
- (D) A and B

CODING-DECODING (Q38-Q40)

Q38. In a certain code, COMPUTER is written as RETUPMOC. How will KEYBOARD be written in that code?

- (A) KEYBAORD
- (B) DRAOBYEK
- (C) BOARDKEY
- (D) YDRAOBEK

Q39. In a certain code language, the value of a word is the sum of the positional values of its letters (A=1, B=2, ... Z=26). PINK = 50, BLUE = 40, RED = 27. What is the code for GREEN?

- (A) 45
- (B) 49
- (C) 52
- (D) 55

Q40. In a code, each letter is shifted 2 places forward (A→C, B→D, ..., Y→A, Z→B). How is MANGO coded?

- (A) OCPIQ
- (B) NBOIP
- (C) PCQJR
- (D) OCLPQ

SECTION 3: VERBAL ABILITY (Q41-Q60)

Topics: Reading Comprehension | Error Detection | Vocabulary | Fill in the Blanks

READING COMPREHENSION (Q41-Q45)

Read the following passage and answer Q41-Q45:

Artificial intelligence has emerged as one of the most transformative technologies of the 21st century. Its applications span across healthcare, finance, education, and transportation. While AI promises to increase efficiency and reduce human error, critics warn of potential job displacement and ethical concerns regarding data privacy. Many experts believe that rather than replacing humans, AI will augment human capabilities, creating new opportunities while necessitating workforce adaptation. Governments worldwide are now grappling with the need to create regulatory frameworks that balance innovation with the protection of citizen rights.

Q41. What is the primary concern raised by critics regarding AI?

- (A) Lack of innovation
- (B) Job displacement and ethical issues
- (C) Government interference
- (D) Limited healthcare applications

Q42. According to the passage, what do many experts believe AI will do to human capabilities?

- (A) Replace them completely
- (B) Diminish them gradually
- (C) Augment them
- (D) Have no significant effect

Q43. What are governments worldwide currently dealing with, as per the passage?

- (A) Funding AI research
- (B) Creating regulatory frameworks
- (C) Banning AI applications
- (D) Developing new AI hardware

Q44. Which sector is NOT mentioned in the passage as an application area of AI?

- (A) Healthcare
- (B) Agriculture
- (C) Finance
- (D) Transportation

Q45. The word "augment" in the passage most nearly means:

- (A) Replace
- (B) Diminish
- (C) Enhance
- (D) Eliminate

GRAMMAR & ERROR DETECTION (Q46-Q50)

Directions (Q46-48): Identify the sentence that contains a grammatical error.

Q46. Identify the sentence that contains a grammatical error.

- (A) She has been working here for five years.

- (B) He don't know the answer to the question.
- (C) They arrived at the station on time.
- (D) We finished the project last night.

Q47. Identify the sentence that contains a grammatical error.

- (A) Neither of the two options are correct.
- (B) The books on the shelves belong to my teacher.
- (C) Each student has submitted their assignment.
- (D) The committee has made its final decision.

Q48. Identify the sentence that contains a grammatical error.

- (A) He has been working since morning.
- (B) I prefer coffee than tea.
- (C) She is one of the best students in the class.
- (D) They have successfully completed the task.

Q49. Choose the correctly spelled word:

- (A) Accomodation
- (B) Occurence
- (C) Accommodation
- (D) Acommodation

Q50. Which sentence uses the word correctly?

- (A) Less people came to the meeting.
- (B) Fewer people came to the meeting.
- (C) The amount of students in the class increased.
- (D) She has less friends than him.

VOCABULARY — SYNONYMS & ANTONYMS (Q51-Q55)

Q51. Choose the synonym of EPHEMERAL:

- (A) Eternal
- (B) Transient
- (C) Ancient
- (D) Enduring

Q52. Choose the synonym of VERBOSE:

- (A) Concise
- (B) Wordy
- (C) Silent
- (D) Precise

Q53. Choose the synonym of BENEVOLENT:

- (A) Hostile
- (B) Indifferent
- (C) Charitable
- (D) Wicked

Q54. Choose the synonym of AMBIGUOUS:

- (A) Clear
- (B) Definite
- (C) Vague
- (D) Precise

Q55. Choose the ANTONYM of LOQUACIOUS:

- (A) Talkative
- (B) Garrulous
- (C) Taciturn
- (D) Verbose

FILL IN THE BLANKS (Q56-Q60)

Q56. Despite his best efforts, he failed to _____ the audience with his speech.

- (A) Bore
- (B) Captivate
- (C) Confuse
- (D) Ignore

Q57. The scientist's discovery was so significant that it _____ the entire field of biology.

- (A) Deteriorated
- (B) Revolutionized
- (C) Complicated
- (D) Simplified

Q58. He spoke so _____ that everyone in the room was convinced by his argument.

- (A) Barely
- (B) Incoherently
- (C) Persuasively
- (D) Timidly

Q59. The two countries signed a _____ agreement to promote trade between them.

- (A) Bilateral
- (B) Unilateral
- (C) Multilateral
- (D) Peripheral

Q60. Choose the correct passive voice form of: "They are building a new hospital."

- (A) A new hospital was built by them.
- (B) A new hospital is being built by them.
- (C) A new hospital will be built by them.
- (D) A new hospital has been built by them.

ANSWER KEY

Q. No.	Ans	Q. No.	Ans	Q. No.	Ans	Q. No.	Ans
Q1	B	Q16	C	Q31	D	Q46	B
Q2	A	Q17	C	Q32	B	Q47	A
Q3	C	Q18	B	Q33	D	Q48	B
Q4	B	Q19	A	Q34	C	Q49	C
Q5	D	Q20	C	Q35	B	Q50	B
Q6	C	Q21	B	Q36	B	Q51	B
Q7	B	Q22	B	Q37	D	Q52	B
Q8	A	Q23	A	Q38	B	Q53	C
Q9	C	Q24	C	Q39	B	Q54	C
Q10	B	Q25	B	Q40	A	Q55	C
Q11	C	Q26	A	Q41	B	Q56	B
Q12	A	Q27	B	Q42	C	Q57	B
Q13	C	Q28	C	Q43	B	Q58	C
Q14	A	Q29	A	Q44	B	Q59	A
Q15	A	Q30	A	Q45	C	Q60	B

DETAILED SOLUTIONS

SECTION 1: QUANTITATIVE APTITUDE SOLUTIONS

Q1. Ans: (B) Rs. 7,040

Weighted investments: $A = 5000 \times 12 = 60,000$ | $B = 8000 \times 6 = 48,000$ | $C = 3000 \times 9 = 27,000$. Ratio = $60:48:27 = 20:16:9$. Total parts = 45. B's share = $(16/45) \times 19,800 = \text{Rs. } 7,040$.

Q2. Ans: (A) Rs. 4,500

Profit-sharing ratio = (Capital x Time). $A : B = (4 \times 8) : (5 \times 6) = 32 : 30 = 16 : 15$. If A's profit = Rs. 4,800 for 16 parts, then 1 part = 300. B's profit = $15 \times 300 = \text{Rs. } 4,500$.

Q3. Ans: (C) 60 litres

Milk = $(2/3) \times 60 = 40\text{L}$, Water = 20L. For ratio 1:2, water must be twice milk. Water needed = $2 \times 40 = 80\text{L}$. Additional water = $80 - 20 = 60$ litres.

Q4. Ans: (B) Rs. 53,760

A invests from month 1 to 12: $40,000 \times 12 = 4,80,000$. B invests from month 4 to 12 (9 months): $60,000 \times 9 = 5,40,000$. C invests from month 7 to 12 (6 months): $80,000 \times 6 = 4,80,000$. Ratio = $480 : 540 : 480 = 8 : 9 : 8$. Total parts = 25. C's share = $(8/25) \times 1,68,000 = \text{Rs. } 53,760$.

Q5. Ans: (D) 55

Let the numbers be $3x$ and $5x$. After subtracting 9: $(3x-9)/(5x-9) = 12/23$. $23(3x-9) = 12(5x-9) \Rightarrow 69x - 207 = 60x - 108 \Rightarrow 9x = 99 \Rightarrow x = 11$. Larger number = $5 \times 11 = 55$.

Q6. Ans: (C) 45 km/h

Relative speed of train with respect to man = $150/15 = 10 \text{ m/s} = 36 \text{ km/h}$. Since both move in the same direction, Train speed = Relative speed + Man's speed = $36 + 9 = 45 \text{ km/h}$.

Q7. Ans: (B) 12 seconds

Trains run in opposite directions, so relative speed = $50 + 40 = 90 \text{ km/h} = 90 \times (5/18) = 25 \text{ m/s}$. Total length = $120 + 180 = 300 \text{ m}$. Time = $300/25 = 12$ seconds.

Q8. Ans: (A) 4.8 km/h

When same distance is covered at two different speeds, use harmonic mean formula: Average speed = $2ab/(a+b) = (2 \times 4 \times 6)/(4 + 6) = 48/10 = 4.8 \text{ km/h}$.

Q9. Ans: (C) 48 km/h

Let total distance = $2d$. Time for first half = $d/40$. Time for second half = $d/60$. Total time = $d/40 + d/60 = (3d + 2d)/120 = 5d/120$. Average speed = $2d / (5d/120) = 240/5 = 48 \text{ km/h}$.

Q10. Ans: (B) 5:00 PM, 120 km

Head start of thief = 1 hour $\times$ 40 km/h = 40 km. Relative speed (police - thief) = 60 - 40 = 20 km/h. Time to catch = 40/20 = 2 hours after police start (3 PM) = 5:00 PM. Distance from start = 60 $\times$ 2 = 120 km.

Q11. Ans: (C) 198

Given $x + 1/x = 6$. Step 1: $x^2 + 1/x^2 = (x + 1/x)^2 - 2 = 36 - 2 = 34$. Step 2: $x^3 + 1/x^3 = (x + 1/x)(x^2 - 1 + 1/x^2) = 6 \times (34 - 1) = 6 \times 33 = 198$.

Q12. Ans: (A) Sum = 8/3, Product = 5/3

For equation $ax^2 + bx + c = 0$: Sum of roots = $-b/a$, Product = c/a . Here $a=3$, $b=-8$, $c=5$. Sum = $-(-8)/3 = 8/3$. Product = $5/3$.

Q13. Ans: (C) 9

For equal roots, discriminant = 0: $b^2 - 4ac = 0$. Here $a=k$, $b=-6$, $c=1$. $(-6)^2 - 4(k)(1) = 0 \Rightarrow 36 - 4k = 0 \Rightarrow k = 9$.

Q14. Ans: (A) $x^2 - 10x + 6 = 0$

Error-in-roots rule: Person who copied constant wrong $\Rightarrow$ sum of roots is CORRECT. Sum = $2 + 8 = 10$. Person who copied x-coefficient wrong $\Rightarrow$ product of roots is CORRECT. Product = $1 \times 6 = 6$. Correct equation: $x^2 - 10x + 6 = 0$.

Q15. Ans: (A) $x^2 - 12x + 35 = 0$

Roots of $x^2 - 5x + 6 = 0$: $\alpha + \beta = 5$, $\alpha\beta = 6$. New roots: $2\alpha+1$ and $2\beta+1$. New sum = $2(\alpha+\beta) + 2 = 10 + 2 = 12$. New product = $(2\alpha+1)(2\beta+1) = 4\alpha\beta + 2(\alpha+\beta) + 1 = 24 + 10 + 1 = 35$. Equation: $x^2 - 12x + 35 = 0$.

Q16. Ans: (C) 184 m²

Inner path area = Total area - Inner rectangle area. Width of path = 2m (inside). Inner dimensions: $(30-4) \times (20-4) = 26 \times 16$. Path area = $(30 \times 20) - (26 \times 16) = 600 - 416 = 184$ m².

Q17. Ans: (C) 27

Volume of large sphere = $(4/3)\pi r^3 = (4/3)\pi(6)^3 = 288\pi$ cm³. Volume of small sphere = $(4/3)\pi(2)^3 = (32/3)\pi$ cm³. Number of small spheres = $288\pi \div (32\pi/3) = 288 \times 3/32 = 27$.

Q18. Ans: (B) 84 sq cm

Sides $a=13$, $b=14$, $c=15$. Semi-perimeter $s = (13+14+15)/2 = 21$. Area = $\sqrt{[s(s-a)(s-b)(s-c)]} = \sqrt{[21 \times 8 \times 7 \times 6]} = \sqrt{7056} = 84$ sq cm.

Q19. Ans: (A) 154 cm³

For same base and height: Volume of cone = $(1/3) \times$ Volume of cylinder. Cone volume = $(1/3) \times 462 = 154$ cm³.

Q20. Ans: (C) 8% increase

Percentage change in area = $P + Q + \frac{PQ}{100}$, where $P = +20\%$ (increase) and $Q = -10\%$ (decrease). $= 20 + (-10) + \frac{(20 \times (-10))}{100} = 20 - 10 - 2 = 8\%$ increase.

SECTION 2: REASONING SOLUTIONS

Q21. Ans: (B) Sister

Only son of Raj's grandfather = Raj's father. Daughter of Raj's father = Raj's sister. So the woman is Raj's SISTER.

Q22. Ans: (B) Grandson

E is D's wife. D is C's father (C is A's mother). So D is A's maternal grandfather. E (D's wife) is A's maternal grandmother. Therefore A is E's GRANDSON.

Q23. Ans: (A) Mother

"Only daughter of my mother" = the woman herself. So the man's mother = the woman. The woman is the man's MOTHER.

Q24. Ans: (C) Father

A + B: A is father of B. B - C: B is sister of C (so B and C are siblings). C x D: C is brother of D (so C and D are siblings). Since A is father of B, and B, C, D are all siblings, A is also father of C and D. A is FATHER of D.

Q25. Ans: (B) 3 km East

From P: Walk 5km North (reach N), turn East walk 3km (reach E), turn South walk 5km (reach S). Net North-South displacement = 5 - 5 = 0. Net East-West = 3 km East. Distance from P = 3 km East.

Q26. Ans: (A) 3 km North

4 km East, then 3 km North, then 4 km West. East and West cancel out (4 - 4 = 0). Net displacement = 3 km North.

Q27. Ans: (B) 4 km West

6 km South, then turn right (now facing West) walk 4 km, then turn right (now facing North) walk 6 km. North-South: -6 + 6 = 0. East-West: -4 km (West). She is 4 km West of office.

Q28. Ans: (C) 8 km

A's position: 5km North then 4km East = coordinates (4, 5). B's position: 4km East then 3km South = coordinates (4, -3). Distance = $\sqrt{[(4-4)^2 + (5-(-3))^2]} = \sqrt{[0 + 64]} = 8$ km.

Q29. Ans: (A) Only I follows

All birds are animals (A→B) + No animal is a plant (E-type, symmetric: No plant is animal). Since all birds are animals, and no animal is a plant, no bird can be a plant. Conclusion I follows. Conclusion II (some birds are plants) contradicts I. Only I follows.

Q30. Ans: (A) Only I follows

Some books are pens (I-type) + All pens are pencils (A-type). By syllogism: Some books are pencils (I follows). "All pencils are books" cannot be concluded (we only know SOME books, not ALL, are pens). Only Conclusion I follows.

Q31. Ans: (D) Neither conclusion follows

All trees are flowers + Some flowers are fruits. The "some flowers" that are fruits may NOT be the same flowers that are trees. No direct link between trees and fruits can be established. Neither Conclusion I nor II necessarily follows.

Q32. Ans: (B) Only II follows

No cat is a dog (E-type, symmetric: No dog is a cat). Some dogs are animals. These animals (which are dogs) are NOT cats. So some animals are not cats => Conclusion II follows. Conclusion I (No cat is an animal) is too strong; cannot be concluded. Only II follows.

Q33. Ans: (D) All I, II and III follow

All mobiles are phones + All phones are gadgets => All mobiles are gadgets (I: A+A=A). Some gadgets are electronics => by conversion, Some electronics are gadgets (II follows). All mobiles are gadgets => by partial conversion, Some gadgets are mobiles (III follows). All three conclusions follow.

Q34. Ans: (C) D

Arrangement from left to right: A(1) - F(2) - C(3) - D(4) - E(5) - B(6). C is at position 3. The person immediately to the right of C is at position 4 = D.

Q35. Ans: (B) C

Arrangement: A(1) - F(2) - C(3) - D(4) - E(5) - B(6). 4th from the right: B(1st), E(2nd), D(3rd), C(4th). Answer: C.

Q36. Ans: (B) 3

Arrangement: A(1) - F(2) - C(3) - D(4) - E(5) - B(6). A is at position 1, E is at position 5. Between them: F(2), C(3), D(4) = 3 persons.

Q37. Ans: (D) A and B

Arrangement: A(1) - F(2) - C(3) - D(4) - E(5) - B(6). A is at the leftmost end (position 1) and B is at the rightmost end (position 6).

Q38. Ans: (B) DRAOBYEK

The code reverses the entire word. COMPUTER reversed = RETUPMOC. Similarly, KEYBOARD reversed = DRAOBYEK.

Q39. Ans: (B) 49

Value = sum of letter positions. GREEN: G=7, R=18, E=5, E=5, N=14. Sum = 7+18+5+5+14 = 49. Verification: PINK=16+9+14+11=50, BLUE=2+12+21+5=40, RED=18+5+4=27. Confirmed.

Q40. Ans: (A) OCPIQ

Each letter shifts 2 places forward: M→O, A→C, N→P, G→I, O→Q. MANGO = OCPIQ.

SECTION 3: VERBAL ABILITY SOLUTIONS

Q41. Ans: (B) Job displacement and ethical issues

The passage states: "critics warn of potential job displacement and ethical concerns regarding data privacy." Option B correctly identifies both concerns.

Q42. Ans: (C) Augment them

The passage explicitly states: "rather than replacing humans, AI will augment human capabilities." Augment means to enhance or supplement.

Q43. Ans: (B) Creating regulatory frameworks

The passage states: "Governments worldwide are now grappling with the need to create regulatory frameworks that balance innovation with the protection of citizen rights."

Q44. Ans: (B) Agriculture

The passage mentions healthcare, finance, education, and transportation as AI application areas. Agriculture is NOT mentioned in the passage.

Q45. Ans: (C) Enhance

"Augment" means to make something greater by adding to it; to supplement or enhance. In this context, AI will enhance (improve/add to) human capabilities.

Q46. Ans: (B) "He don't know the answer..."

Error: "don't" should be "doesn't." With third-person singular subjects (he/she/it), use "does not" (doesn't), not "do not" (don't). Correct: "He doesn't know the answer."

Q47. Ans: (A) "Neither of the two options are correct."

Error: "are" should be "is." "Neither" is singular and takes a singular verb. Correct: "Neither of the two options is correct."

Q48. Ans: (B) "I prefer coffee than tea."

Error: "than" should be "to." The correct construction with "prefer" is: "prefer X to Y" (not "than"). Correct: "I prefer coffee to tea."

Q49. Ans: (C) Accommodation

Correct spelling: Accommodation (double c, double m). Common misspellings: Accomodation (single m), Occurence (should be Occurrence).

Q50. Ans: (B) "Fewer people came to the meeting."

Rule: Use "fewer" for countable nouns and "less" for uncountable/mass nouns. "People" is countable, so "fewer" is correct. "Less people" and "less friends" are grammatically incorrect.

Q51. Ans: (B) Transient

Ephemeral means lasting for a very short time. Synonym: Transient (lasting only a short time). Antonyms would be: Eternal, Enduring, Lasting.

Q52. Ans: (B) Wordy

Verbose means using more words than necessary; long-winded. Synonym: Wordy. Antonyms: Concise, Succinct, Brief.

Q53. Ans: (C) Charitable

Benevolent means well-meaning and kind; generous. Synonym: Charitable. Antonyms: Hostile, Wicked, Malevolent.

Q54. Ans: (C) Vague

Ambiguous means open to more than one interpretation; unclear. Synonym: Vague. Antonyms: Clear, Definite, Precise, Unambiguous.

Q55. Ans: (C) Taciturn

Loquacious means tending to talk a great deal; talkative. Its ANTONYM is Taciturn (reserved; saying little). Talkative, Garrulous, Verbose are all synonyms of Loquacious.

Q56. Ans: (B) Captivate

"Failed to captivate" means he could not attract or hold the audience's attention. The contrast implied by "despite his best efforts" suggests the audience was not engaged, making "captivate" (to attract and hold interest) the most appropriate word.

Q57. Ans: (B) Revolutionized

"Revolutionized" means caused a complete change in something. A discovery so significant as to transform an entire field would "revolutionize" it. "Simplified" and "Deteriorated" don't match the positive, transformative context.

Q58. Ans: (C) Persuasively

The result "everyone was convinced" tells us the manner of speaking led to conviction. "Persuasively" (in a way intended to make someone believe something) is the correct word. "Barely" and "timidly" would not result in convincing others.

Q59. Ans: (A) Bilateral

"Bilateral" means involving two parties — perfectly fitting "two countries." "Unilateral" = one party. "Multilateral" = many parties. "Peripheral" = relating to the edge/margin (unrelated).

Q60. Ans: (B) "A new hospital is being built by them."

Active: "They are building a new hospital." (Present Continuous). Passive: Object (a new hospital) + is/am/are + being + V3 + by + Subject. = "A new hospital is being built by them." Option A uses past tense (was built), C uses future (will be built), D uses perfect (has been).